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BS PROJECT

Implementation of Molecular Dynamics for Active Matter: A Study of the Lennard-Jones Liquid and the Vicsek Model

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Abstract
Active matter systems comprise self-propelled agents that convert energy into mechanical motion, leading to emergent collective behaviors such as flocking and clustering. In this study, I develop and optimize a molecular dynamics (MD) framework to simulate both passive Lennard-Jones liquids and active particles governed by the Vicsek model within a 2D rectangular simulation box. To bypass the $O(N^2)$ computational bottleneck of neighbor searches, I implemented a Linked-Cell List algorithm coupled with Just-In-Time (JIT) compilation via the Python Numba library, achieving $O(N)$ scaling. The framework is first validated by computing the Radial Distribution Function of a dense Lennard-Jones fluid. Subsequently, I explore the kinetic phase transitions of the Vicsek model simulating up to 20,480 particles. My results isolate and quantify the emergence of macroscopic global alignment driven by variations in angular noise and particle density, capturing the non-equilibrium continuous phase transition characteristic of active matter within constrained rectangular geometries.

1. Introduction

The study of collective motion in active matter has garnered significant attention across physics, biology, and materials science. Unlike passive particles driven purely by thermal fluctuations and conservative forces, active particles consume energy to propel themselves, keeping the system inherently out of thermodynamic equilibrium. The accurate modeling of such systems requires robust computational frameworks capable of handling tens of thousands of interacting bodies. Traditionally, numerical techniques based on molecular dynamics have been employed to solve the underlying equations governing particle trajectories. By establishing a foundational MD pipeline using standard continuous potentials, I can extend these numerical techniques to active, non-equilibrium models to predict collective macroscopic behavior.

2. Theoretical framework for active matter simulation

2.1. Lennard-Jones Molecular Dynamics

To establish the baseline continuous-space MD framework, I simulate a liquid governed by the Lennard-Jones potential. The system is integrated using a Velocity Verlet algorithm within an NVE ensemble. The structural properties of the liquid are quantified using the Radial Distribution Function, $g(r)$, which describes the probability of finding a particle at a distance r from a reference particle compared to an ideal gas.

2.2. The Vicsek Model

To capture the dynamics of flocking, I utilize the Vicsek model, which provides a minimalist framework for self-propelled particles. I consider a 2D rectangular computational box with dimensions $L_x \times L_y$, subject to periodic boundary conditions, containing N identical particles moving with a constant speed v. The governing equations for the temporal evolution of the i-th particle's position $\vec{r}_i$ and orientation angle θ_i are given by:

$$\vec{r}_i(t + \Delta t) = \vec{r}_i(t) + \vec{v}_i(t)\Delta t \tag{1}$$

$$\theta_i(t + \Delta t) = \langle \theta(t) \rangle_R + \Delta\theta \tag{2}$$

where $\langle \theta(t) \rangle_R$ is the average orientation of all particles within a local interaction radius R, and $\Delta\theta$ is a random scalar noise uniformly distributed in the interval $[-\eta/2, \eta/2]$. The macroscopic order parameter, v_a, is defined as the normalized magnitude of the average velocity vector.

3. Computational setup

Simulating tens of thousands of interacting particles requires careful consideration of algorithmic efficiency. The primary bottleneck in MD is the neighbor-search algorithm. To accelerate convergence and maintain simulation stability, the code architecture heavily leverages the Numba library for JIT compilation.

Specifically, I implemented a Linked-Cell List algorithm to reduce the neighbor-search time complexity from $O(N^2)$ to $O(N)$. The rectangular simulation box is divided into a spatial grid where cell dimensions are bounded by the interaction cutoff radius r_c. Particles only compute interactions with neighbors residing in their own cell or the immediately adjacent cells. This spatial hashing, combined with a precise implementation of the Minimum Image Convention for periodic boundaries across the rectangular domain, ensures the simulation remains highly performant. For the LJ liquid phase, the system utilizes a random overlap-free initialization with a minimum distance threshold, allowing the system to thermalize smoothly into a disordered state without catastrophic overlapping forces.

4. Results and discussion

4.1. Simulation results for Lennard-Jones liquid

The MD framework was first validated by tracking the positions of a 3D system of LJ particles at a density of $\rho = 0.8$. After an initial equilibration phase of 1,000 steps to allow transients to dissipate, the system coordinates were sampled to construct the Radial Distribution Function. The resulting $g(r)$ curve correctly approaches 1.0 at long ranges (the ideal gas limit) and exhibits the characteristic primary and secondary coordination peaks, confirming that the algorithmic pipeline accurately resolves the thermodynamic equilibrium of a dense liquid.

4.2. Simulation results for finite-size scaling and noise

In the active matter regime, the Vicsek model was simulated across varied system sizes (from $N = 80$ to $N = 20,480$) while keeping the overall density strictly constant ($\rho = 4.0$). An annealing strategy was employed, gradually sweeping the noise parameter η downward from 5.0 to 0.0. To combat the

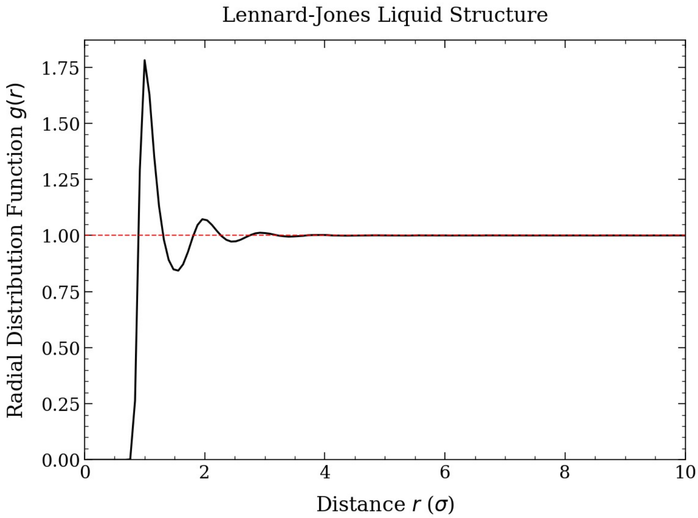

Figure 1. *Radial Distribution Function $g(r)$ for the Lennard-Jones liquid at density $\rho = 0.8$, demonstrating the characteristic primary and secondary coordination shells of a thermalized liquid state..*

phenomenon of critical slowing down near the phase transition, the equilibration time was dynamically scaled relative to the rectangular box dimensions. The results exhibit a clear kinetic phase transition: at high noise, the global order parameter v_a is near zero (disordered state), but as η decreases below a critical threshold, v_a sharply approaches 1.0, indicating the emergence of a highly aligned, macroscopic flock.

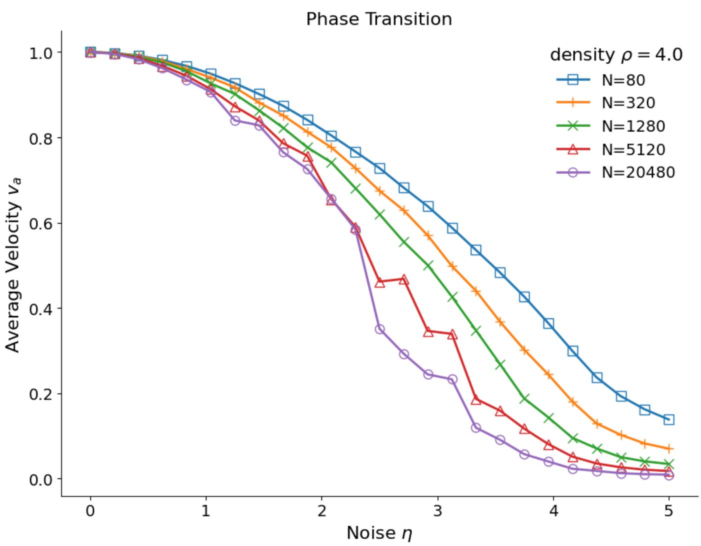

Figure 2. *Kinetic phase transition in the Vicsek model. The average velocity v_a is plotted against angular noise η for various system sizes N at a constant density $\rho = 4.0$..*

4.3. Simulation results for density-driven flocking

To isolate the effect of particle crowding, a secondary experiment locked the noise parameter at $\eta = 2.0$ within a fixed rectangular box of dimensions $L_x = 25.0$ and $L_y = 16.0$ (total area 400). By sweeping the density ρ from 0.1 to 10.0 over multiple ensemble trials, I observed the transition from an uncoordinated active gas to an ordered flock entirely driven by collision frequency. The data confirms that higher particle densities allow the local alignment rules to overcome the inherent angular noise, bridging the gap between microscopic interactions and global order.

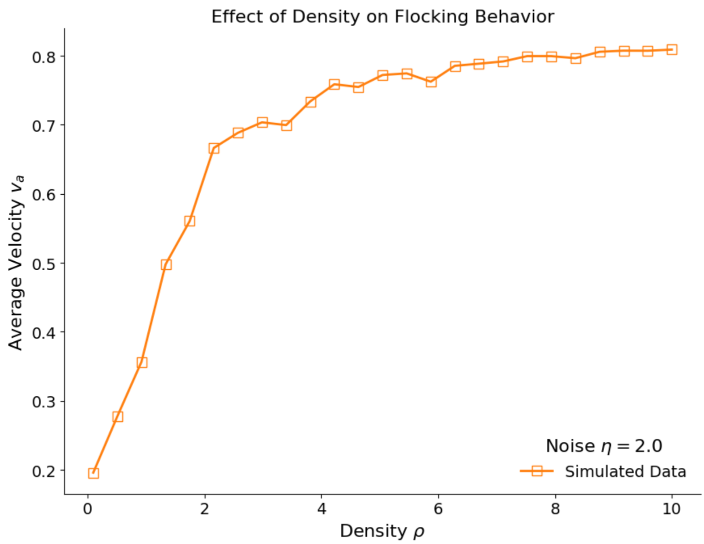

Figure 3. *Effect of particle density on macroscopic flocking behavior. The global order parameter v_a transitions from a disordered gas to an ordered flock as density ρ increases at a constant angular noise $\eta = 2.0$..*

5. Conclusion and future work

In this study, I systematically developed and validated a molecular dynamics framework, applying it to both continuous conservative potentials and discrete active matter models. By optimizing the simulation architecture with Linked-Cell lists and JIT compilation, I successfully modeled the collective dynamics of up to 20,480 particles within a rectangular geometry. The simulations successfully capture the structural distributions of fluids and the complex phase transitions inherent to active flocking systems. Future work will focus on combining these domains by incorporating modified Lennard-Jones interactions into the active model to study the interplay between excluded volume effects, boundary geometries, and active alignment forces.

References

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